SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: **Database Management Systems**
COURSE CODE: **CSA0504**

COURSE OUTCOME COVERED

CO1: *Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships.* (BL1, BL2)

Activity Tool: Concept Mapping (Medium) Assessment
Tool Weightage: 20%

QUESTIONS		
Q.No	**Question**	**Bloom's Level**
1	Construct a concept map linking the following terms: DBMS, Schema, Instance, Data Model, ER Diagram, Entity, Attribute, Relationship.	BL2 – Understand
2	Create a concept map for the three-level ANSI/SPARC architecture showing data independence at each level.	BL2 – Understand
3	Map the relationship between file-based systems and database systems, highlighting the problems solved by DBMS.	BL2 – Understand
4	Design a concept map for ER diagram components: entity, weak entity, attribute types, relationship types, and participation constraints.	BL2 – Understand
5	Construct a concept map comparing Hierarchical, Network, and Relational data models with their key properties.	BL2 – Understand
6	Develop a concept map for the EER model showing how it extends the basic ER model with generalization, specialization, and aggregation.	BL2 – Understand

7	Create a visual concept map for the DBMS software architecture, illustrating the flow from user request to data retrieval.	BL2 – Understand
8	Map the data models discussed in CO1 to real-world applications and explain the suitability of each model.	BL2 – Understand
9	Construct a concept map for schemas and instances, showing how they relate to physical and logical data independence.	BL2 – Understand
10	Design a concept map for the roles and responsibilities in a DBMS environment: DBA, End User, Application Developer.	BL1 – Remember

---

Code of Conduct:

I A i s h w a r y a B / 1 9 2 4 2 1 4 3 2 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

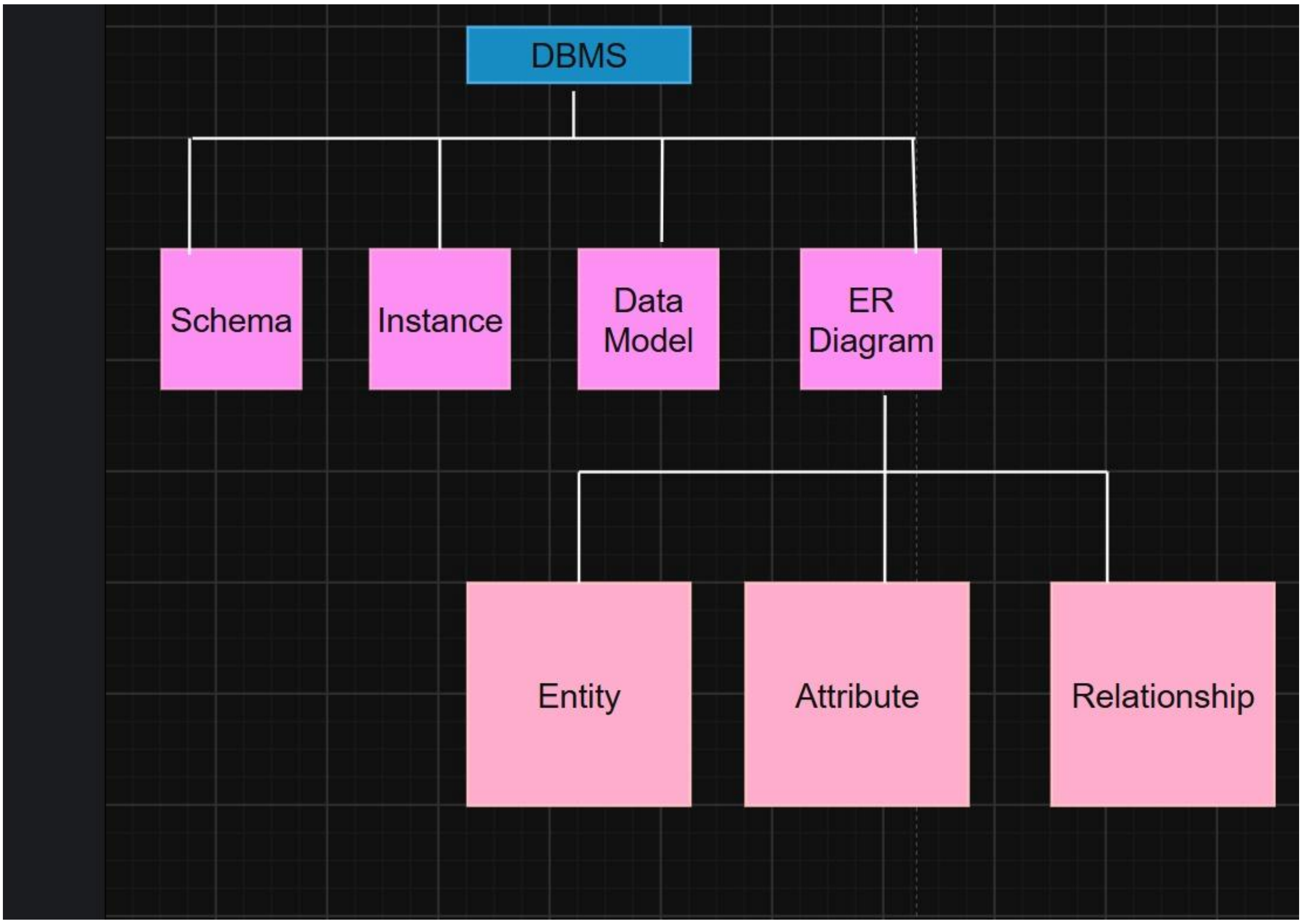
Signature of the student: B Aishwarya

RUBRICS FOR CALCULATION:

Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensivel y	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationship s are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure

Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

1. **Construct a concept map linking the following terms: DBMS, Schema, Instance, Data Model, ER Diagram, Entity, Attribute, Relationship.**



1. DBMS (Database Management System)

DBMS is software that allows users to create, store, retrieve, update, and manage data efficiently. It provides security, consistency, and easy access to data.

Examples: MySQL, Oracle, SQL Server, PostgreSQL.

2. Schema

A Schema is the blueprint or logical design of the database. It defines the structure of tables, columns, relationships, constraints, and data types.

Example:

Student (Student ID, Name, Age, Department)

Key Points:

- Defines the database structure.
- Created before data is stored.
- Rarely changes.

3. Instance

An Instance is the actual data stored in the database at a specific point in time. Unlike the schema, the instance changes whenever records are inserted, updated, or deleted.

Key Points:

- Represents current data.
- Changes frequently.
- Depends on user operations.

4. Data Model

A Data Model is a collection of concepts used to describe how data is organized and how different data items are related.

Common Data Models:

- Hierarchical Model
- Network Model
- Relational Model
- ER Model

Purpose:

- Organizes data.
- Reduces redundancy.
- Improves data consistency.

5. ER Diagram

An Entity-Relationship (ER) Diagram is a graphical representation of the database design. It visually shows entities, their attributes, and the relationships between them.

Purpose:

- Simplifies database design.
- Helps developers understand the database structure.
- Used before creating the actual database.

6. Entity

An Entity is any real-world object or concept for which data is stored.

Examples:

- Student
- Employee
- Customer
- Product

A Student is an entity because information such as ID, name, and age is stored about it.

7. Attribute

An Attribute describes the characteristics or properties of an entity.

Example:

Entity: Student

Attributes:

- Student ID
- Name
- Age
- Department

These attributes provide detailed information about the student entity.

8. Relationship

A **Relationship** connects two or more entities and describes how they interact.

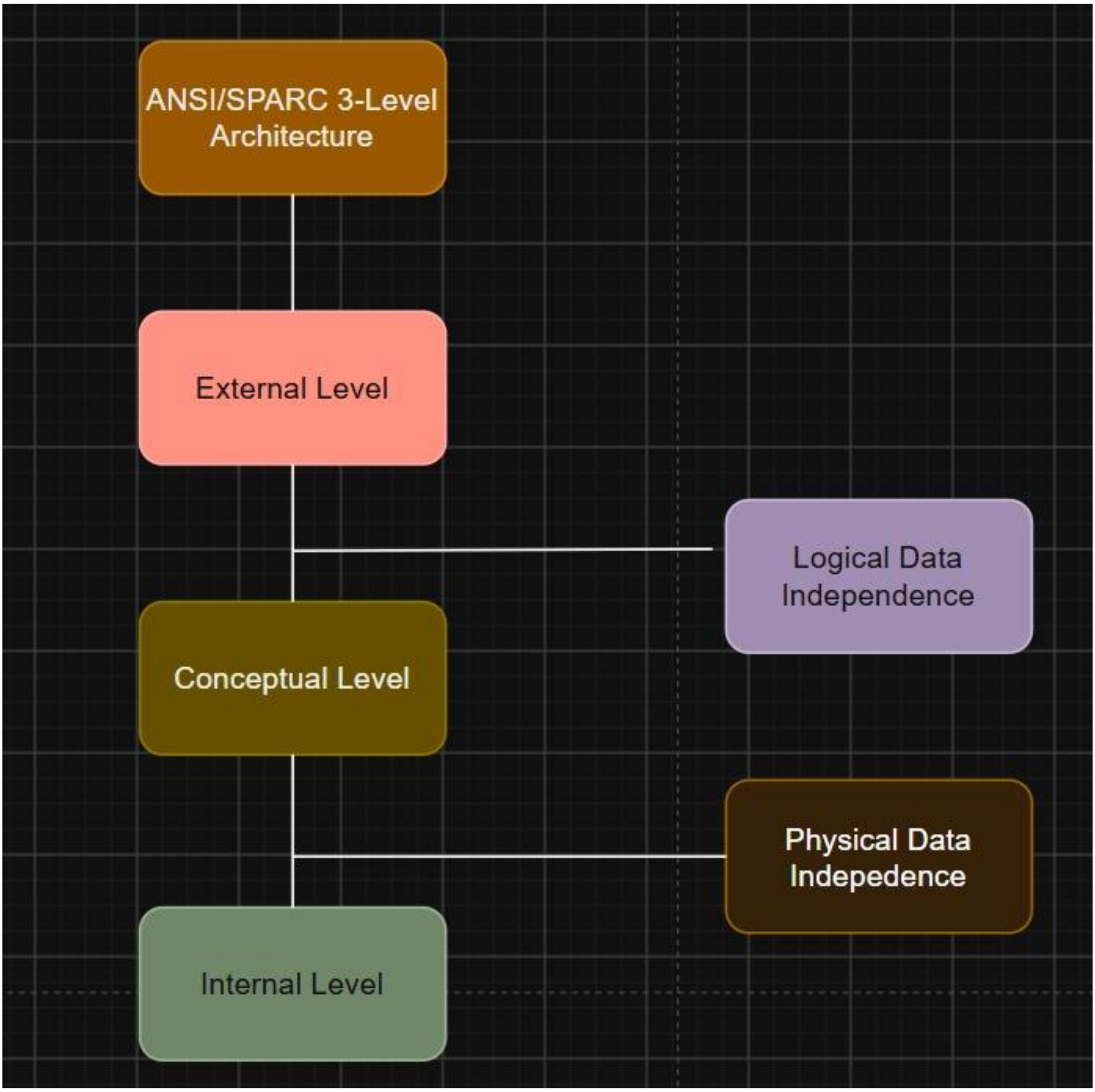
Examples:

- A Student enrols in a Course.
- An Employee works in a department.

Types of Relationships:

- One-to-One (1:1)
- One-to-Many (1:M)
- Many-to-Many (M: N)

2. Create a concept map for the three-level ANSI/SPARC architecture showing data independence at each level



ANSI/SPARC 3-Level Architecture

- A database architecture that separates user views, logical structure, and physical storage.
- It provides data independence by separating different levels of the database.

External Level

- Represents the user's view of the database.
- Each user sees only the data they need.
- Also called the View Level or Subschema.

Conceptual Level

- Represents the overall logical structure of the database.
- Defines entities, relationships, constraints, and data types.
- Also called the Global Schema.

Internal Level

- Represents the physical storage of the database.
- Defines how data is stored in files, indexes, and memory.
- Also called the Physical Level.

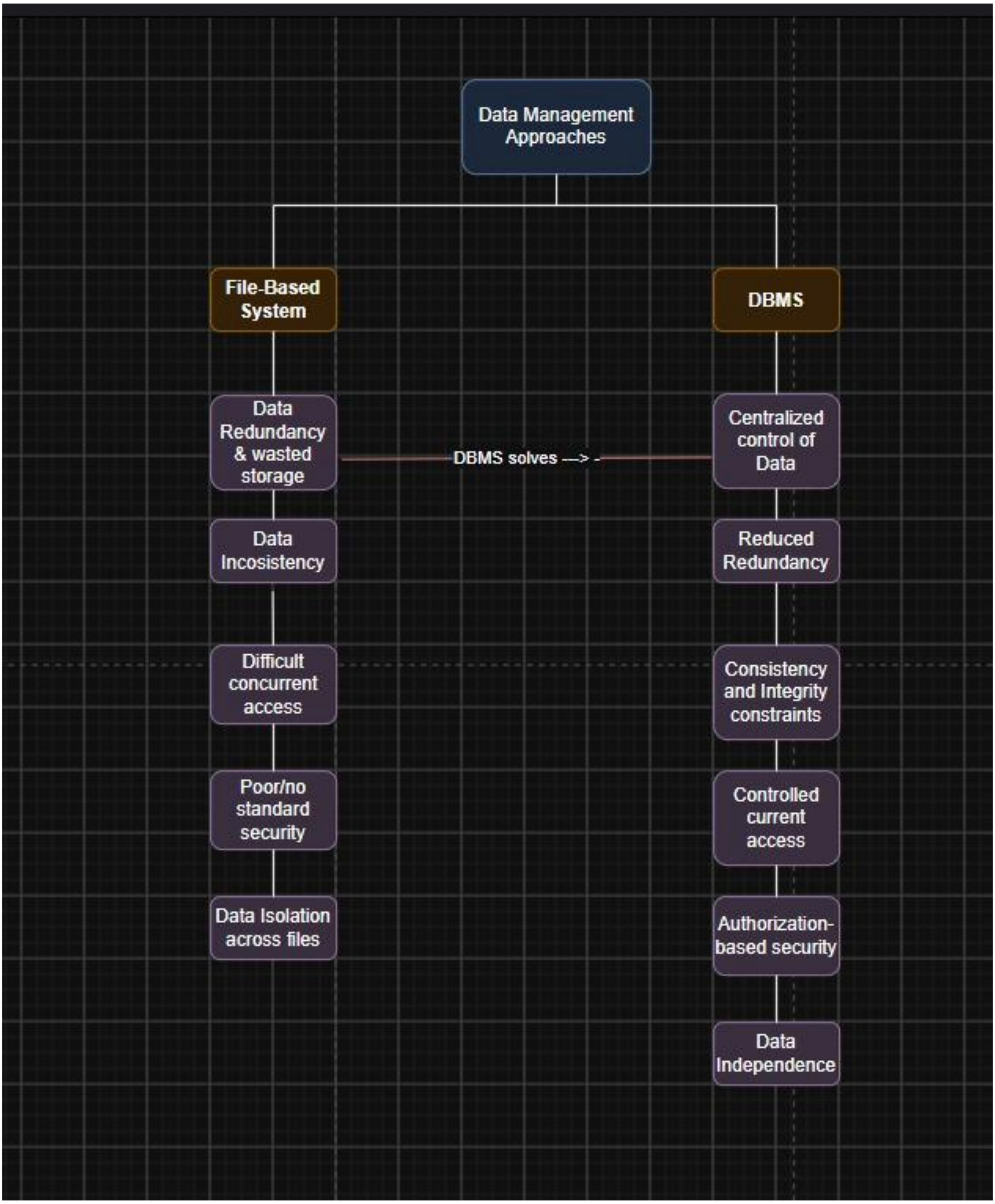
Logical Data Independence

- Ability to modify the Conceptual Level without affecting the External Level.
- User views remain unchanged even if the logical structure changes.

Physical Data Independence

- Ability to modify the Internal Level without affecting the Conceptual Level.
- Changes in storage methods do not affect the database design.

3. Map the relationship between file-based systems and database systems, highlighting the problems solved by DBMS



Data Management Approaches

- Refers to the methods used to store, organize, and manage data.
- The two main approaches are File-Based Systems and Database Management Systems (DBMS).

File-Based System

- Stores data in separate files managed by individual applications.
- Leads to data redundancy and inconsistency.
- Makes concurrent access difficult.
- Provides limited or no standard security.
- Causes data isolation across multiple files.

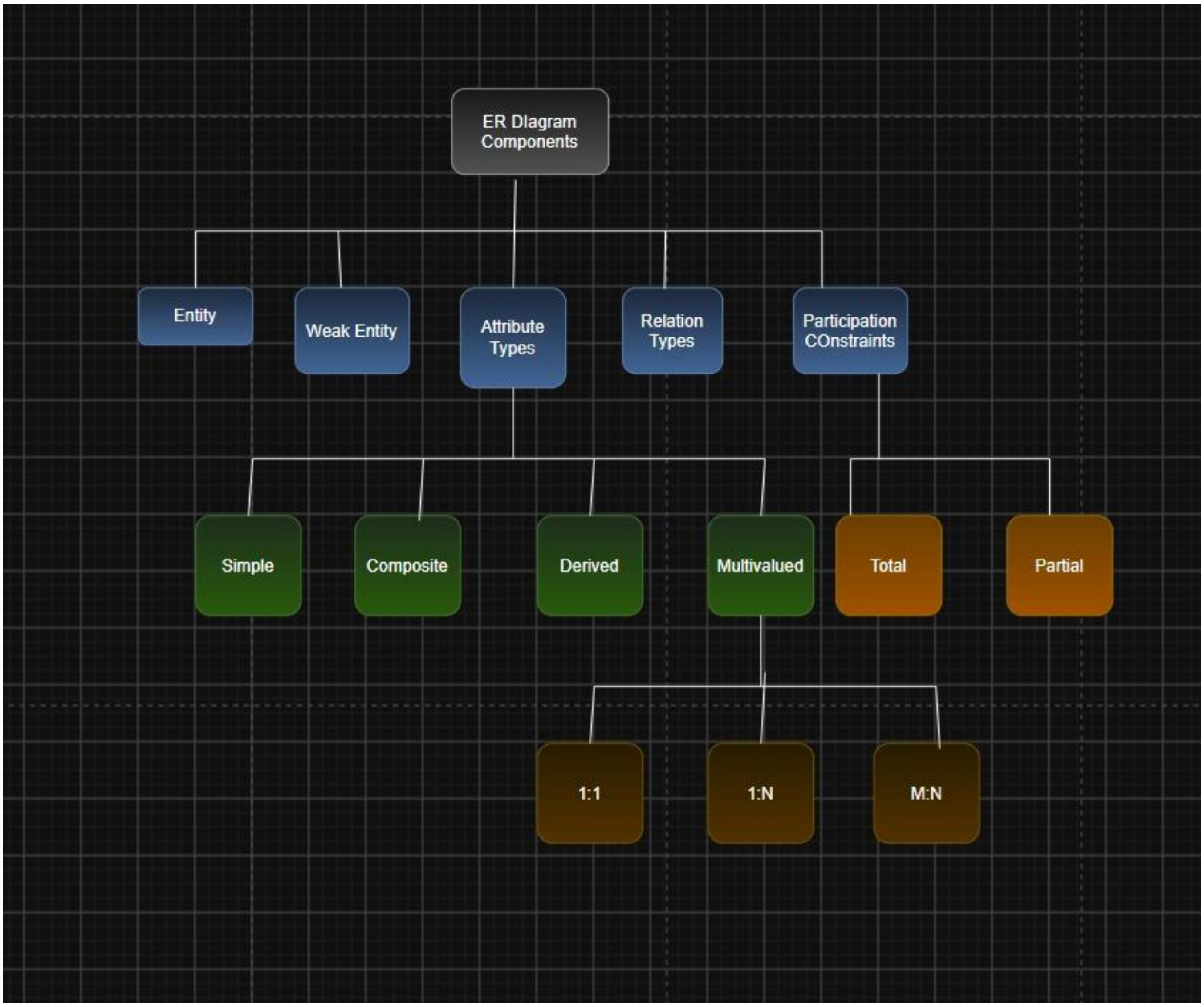
DBMS (Database Management System)

- Stores and manages data in a centralized database.
- Reduces data redundancy and improves consistency.
- Supports controlled concurrent access for multiple users.
- Provides authorization-based security.
- Ensures data independence and efficient data management.

Problems Solved by DBMS

- **Reduces Data Redundancy:** Eliminates duplicate copies of data.
- **Maintains Data Consistency:** Ensures accurate and reliable information.
- **Supports Concurrent Access:** Allows multiple users to access data safely.
- **Provides Better Security:** Protects data using user authentication and permissions.

4. Design a concept map for ER diagram components: entity, weak entity, attribute types, relationship types, and participation constraints.



ER Diagram Components

- ER Diagram components are used to model the structure of a database.
- They include entities, attributes, relationships, and participation constraints.

Entity

- A real-world object or concept about which data is stored.
- Examples: Student, Employee, Customer.

Weak Entity

- An entity that cannot be uniquely identified by its own attributes.
- Depends on a strong entity for identification.
- Example: Dependent of an Employee.

Attribute Types

- **Simple Attribute:** Cannot be divided further (e.g., Age).
- **Composite Attribute:** Can be divided into smaller parts (e.g., Address → Street, City, State).
- **Derived Attribute:** Calculated from other attributes (e.g., Age from Date of Birth).
- **Multivalued Attribute:** Can have multiple values (e.g., Phone Numbers).

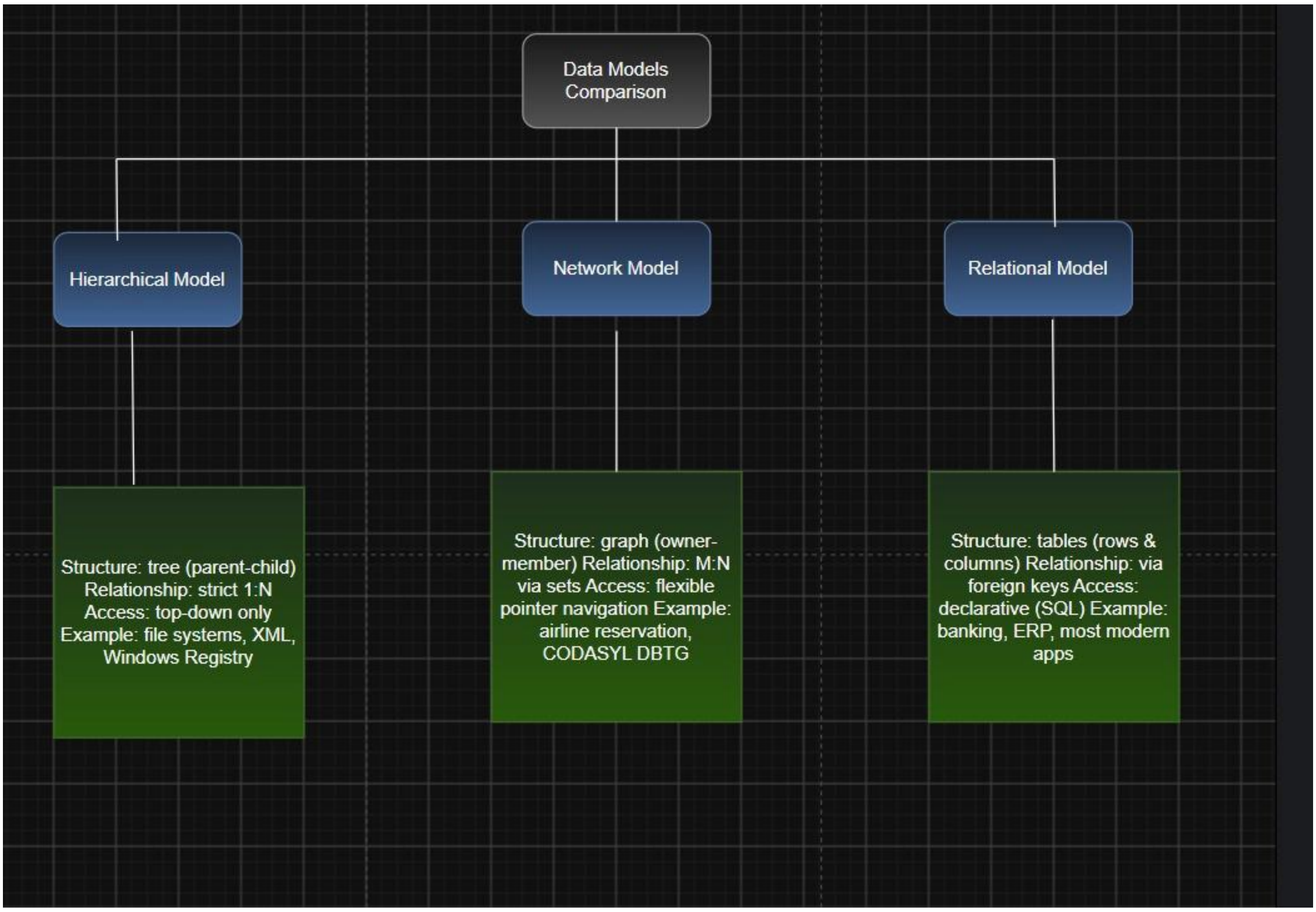
Relationship Types

- Defines how entities are associated with each other.
- **1:1 (One-to-One):** One entity is related to one entity.
- **1: N (One-to-Many):** One entity is related to many entities.
- **M: N (Many-to-Many):** Many entities are related to many entities.

Participation Constraints

- Specifies whether an entity must participate in a relationship.
- **Total Participation:** Every entity must participate.
- **Partial Participation:** Participation is optional.

5. Construct a concept map comparing Hierarchical, Network, and Relational data models with their key properties.



__Data Models Comparison__

- Data models define how data is organized, stored, and accessed in a database.
- The three main data models are Hierarchical, Network, and Relational.

__Hierarchical Model__

- Organizes data in a tree structure (parent-child relationship).
- Each child has only one parent.
- Supports one-to-many (1: N) relationships.
- Examples: File systems, XML, Windows Registry.

__Network Model__

- Organizes data in a graph structure.
- A child can have multiple parents.
- Supports many-to-many (M: N) relationships.

- Examples: Airline reservation systems, CODASYL DBTG.

Relational Model

- Organizes data into tables (rows and columns).
- Relationships are established using primary and foreign keys.
- Data is accessed using SQL (Structured Query Language).

Key Properties

- **Hierarchical Model:** Tree structure, parent-child relationship, top-down access.
- **Network Model:** Graph structure, multiple relationships, flexible navigation.
- **Relational Model:** Table structure, key-based relationships, SQL support.